

Solution Requirements

Date	15 JUNE 2026
Team ID	XXXXXX
Project Name	Sustainable Waste Management Assistant Using Generative AI
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S. No	Requirement Category	Requirement Description	Priority
1	Authentication	Users sign in securely using Firebase Auth; optional Google login; credentials stored safely in Firestore	High
2	Authorization Levels	Two roles – User and Admin. Users can scan and view data; Admins manage collection center info and analytics	High

3	External Interfaces	Integrates Groq API for AI classification, Leaflet.js + OpenStreetMap for maps, and Firebase Firestore for cloud storage	High
4	Transaction Processing	Handles real-time data exchange between frontend and backend via Flask API; stores scan records and retrieves AI responses	High
5	Reporting	Generates interactive charts using Chart.js showing recycling rates, hazardous items, and daily scan frequency	Medium
6	Business Rules	AI must classify each item into one of four categories (recyclable, organic, hazardous, e-waste); save results automatically	High
7	Compliance to Laws or Regulations	Follows data-privacy best practices under GDPR-like principles; no personal data shared with third parties	High
8	Other	Supports location-based filtering for collection centers and eco-friendly recommendations	Medium

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Step 2: Non-Functional Requirements (NFR)

S. No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	System should return AI classification and guidance within 2 seconds of user input	Page loads < 2 s
2	Scalability	Backend and database must handle growth to 10 000 concurrent users without degradation	Supports ≥ 10 000 users
3	Security & Data Privacy	All data transfers encrypted using AES-256; API keys stored in .env files	End-to-end AES-256 encryption
4	Reliability & Availability	Cloud services should maintain 99.9 % uptime with automatic failover	≥ 99.9 % uptime

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5	Usability & Accessibility	Interface must be responsive and accessible on desktop and mobile; complies with WCAG 2.1 AA	WCAG 2.1 AA
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